

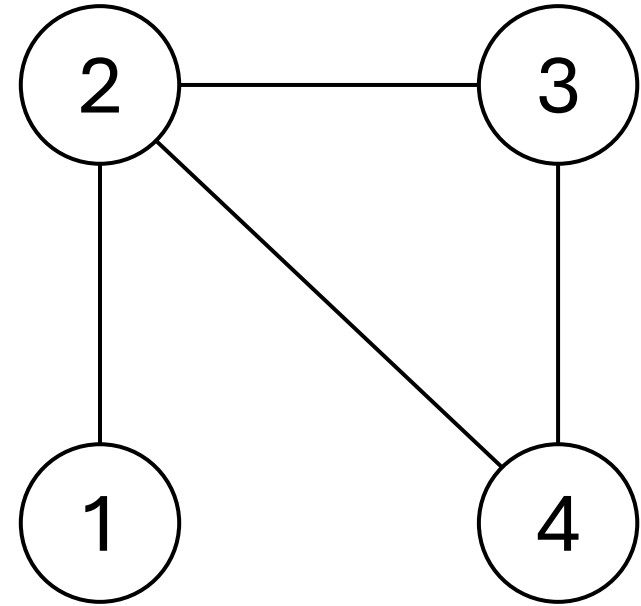


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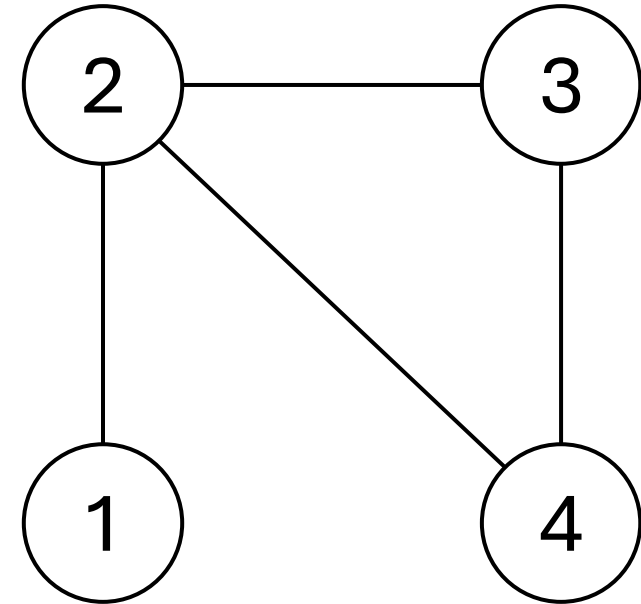
Dynamic matching models under greedy policies: performance paradox and extension to hypergraphs

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- Motivation: kidney paired donation.
- Other applications: book swap, home swap, etc.
- Features:
 1. Compatibility
 2. Random arrivals
 3. Finite waiting time



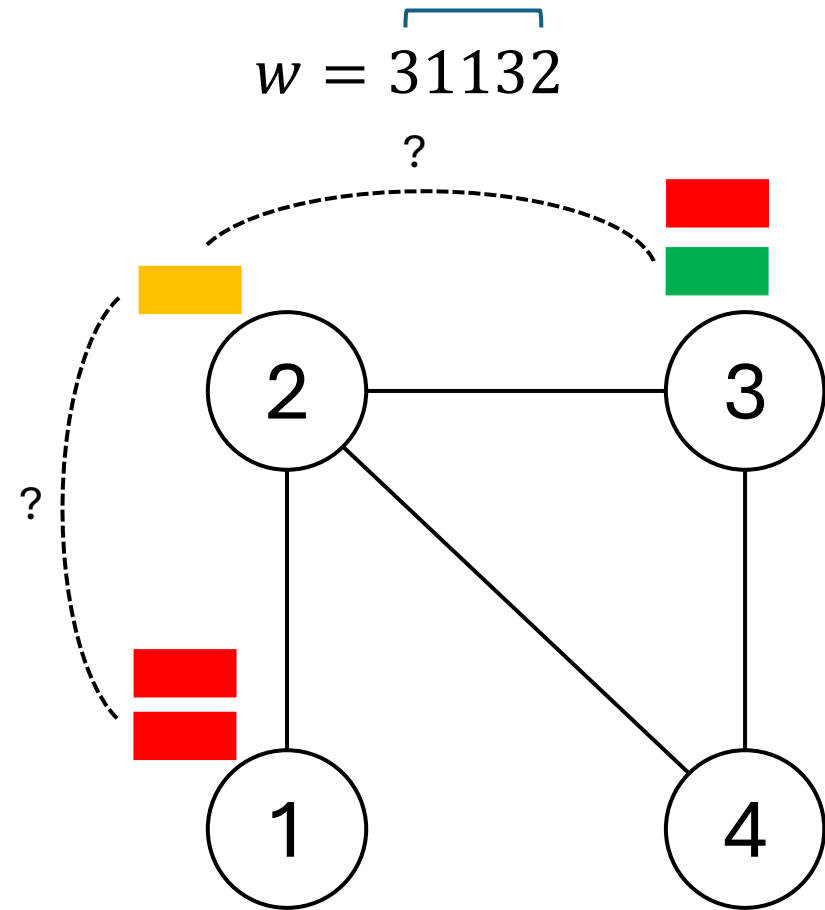
1. Compatibility: Graph $G = (V, E)$
 - Vertices u, v : item classes;
 - Edges $e = (u, v)$: matching types.
Items of class u and v in the buffer *can* form a type- e matching, then leave the buffer.
2. Random arrivals:
 - Distribution μ on V .
 - At time t , a class- v item arrives to the buffer i.i.d. according to μ .
3. Finite waiting time: Stability
 - Under Markovian policies, the buffer is a Markov chain $X(t)$.
 - Stability = X positive recurrent.

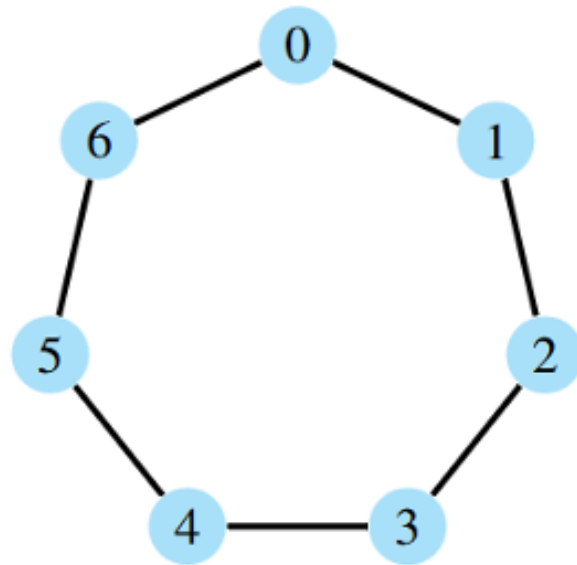


- There can be multiple matchings, and a matching policy is needed.
 - Greedy policies = matching if possible
- First-Come-First-Matched (FCFM)
 - Greedy
 - Always stabilising
 - Perceived as fair
 - Admit a product-form stationary distribution, but very hard to compute.

$$\pi(w) = \frac{1}{\bar{C}} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

$$C = 1 + \sum_{\text{U-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$

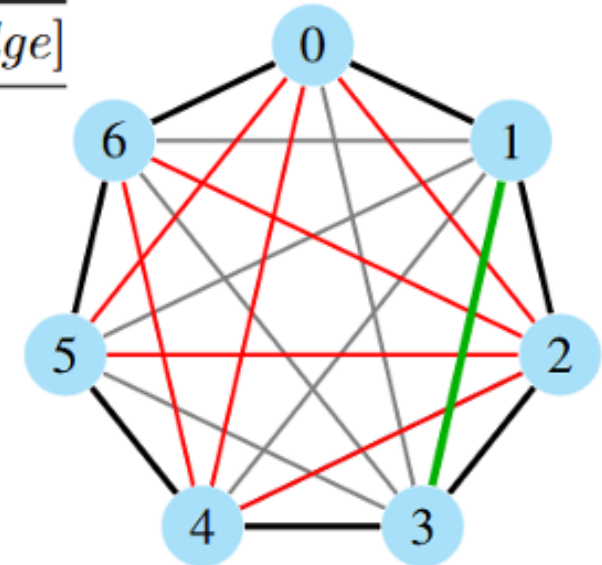




(a)

$Edge$	$\mathbb{E}[G + Edge]$	$Edge$	$\mathbb{E}[G + Edge]$
$\emptyset$	27.8	$\{1, 6\}$	22.5
$\{1, 3\}$	7.8	$\{3, 6\}$	22.5
$\{0, 2\}$	28.8	$\{2, 4\}$	28.4
$\{0, 3\}$	27.6	$\{2, 5\}$	29.2
$\{0, 4\}$	27.9	$\{2, 6\}$	29.2
$\{0, 5\}$	28.7	$\{3, 5\}$	12.8
$\{1, 4\}$	27.1	$\{4, 6\}$	28.9
$\{1, 5\}$	11.9		

(b)



(c)

- Under FCFM policy, performance can worsen if bad edges are added.
- The same holds for *all* greedy policies.

$$C = 1 + \sum_{U\text{-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$

$$\pi(w) = \frac{1}{C} \prod_{i=1}^d \frac{\mu(w_i)}{\mu(N(\{w_1, \dots, w_i\}))}$$

$$C = 1 + \sum_{U\text{-independent}} \sum_{\sigma \in \mathcal{S}(|U|)} \prod_{j=1}^{|U|} \frac{\mu(u_{\sigma(j)})}{\mu(N(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})) - \mu(\{u_{\sigma(1)}, \dots, u_{\sigma(j)}\})}$$

> 0?

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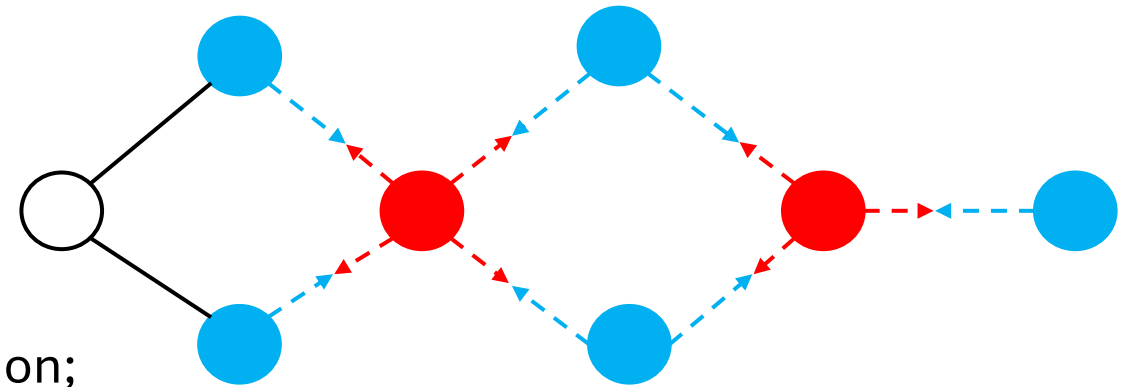
> 0?

Stability criterion (Mairesse & Moyal, 2016):

(G, μ) is stabilized by some policy iff for all independent sets U ,

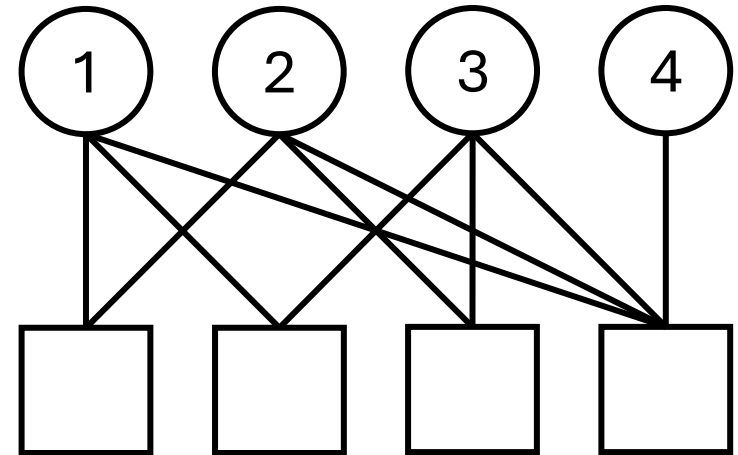
$$\mu(U) < \mu(N(U))$$

- Necessity: law of large number
- Sufficiency:
 - negative drift for quadratic Lyapunov function;
 - First-Come-First-Matched, Match-the-Longest.

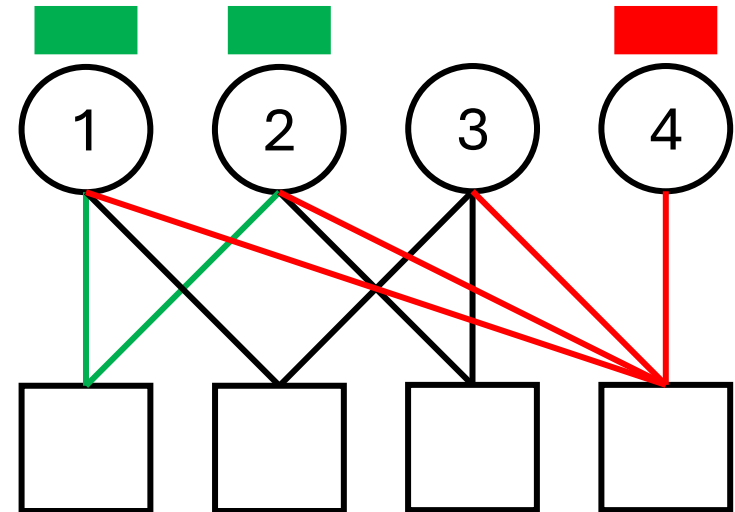


What about *hypergraphs*?

- A **hypergraph** $G = (V, E)$:
 - Vertices: item classes $\bigcirc$;
 - Edges: matching types $\square$.
- Distribution μ on V .
- At time t , a class- v item arrives to the buffer i.i.d. according to μ .
- If **one item for each class u_i** are in the buffer, and $\{u_1, \dots, u_k\} \in E$, they *can* be combined and form a type- e matching, then leave the buffer together.
- Motivation: multi-party kidney donation.
 - Three-party donation decreases waiting time.
- Other applications: multi-stop ride hailing, multi-party good exchange.



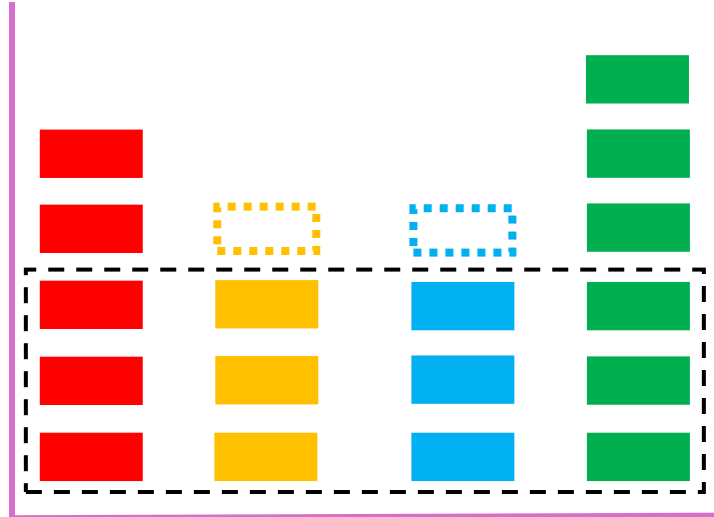
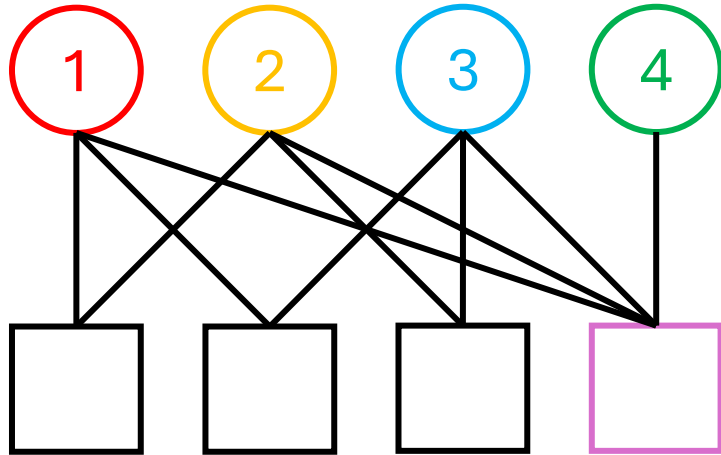
- But greedy policies are *not* maximally stable.
- Problem: given (G, μ) ,
 - What are the necessary and sufficient criteria for stability?
 - If it is stabilisable, find a stabilising policy.
- Question: what is a good alternative to greedy policies?



1. Online assignment rate: $\nu = (\nu_{u,e})_{u \in e, e \in E}$
2. Online assignment policy:
 - At time t , choose a rate ν .
 - If vertex u is drawn, assign the newly arrived item to type e with probability $\frac{\nu_{u,e}}{\mu_u}$.
 - **This item only leaves the buffer via a type- e matching.**
3. A support U is $U = \{(u, e) | q_{u,e} > 0\}$ for some buffer state $q = (q_{u,e})_{u \in e, e \in E}$.

Stability criterion (N., Bušić, 2026): (G, μ) is stabilizable iff **for all supports U** , there exists an online assignment rate ν^U (depending on U) such that for all $(v, e) \in U$,

$$\boxed{\nu_{v,e}^U < \frac{1}{|e|} \sum_{u \in e} \nu_{u,e}^U}$$



Example:

- Support $U \supseteq \{(1, e), (4, e)\}$.
- To have a matching, we need a class-2 and a class-3 item.
- We need them to arrive more often than class-1 and class-4 items to balance out the buffer.
- The stability criterion now reads:

$$\lambda_{e, \blacksquare} < \frac{\lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare}}{4}$$

and

$$\lambda_{e, \blacksquare} < \frac{\lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare} + \lambda_{e, \blacksquare}}{4}$$

- How to quantify the imbalance?

$$\mathcal{L} = (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + (\blacksquare_e - \blacksquare_e)^2 + \dots$$

- **Sufficiency:**

- Lyapunov function: $\mathcal{L}(X(t)) = \sum_{e \in E} \sum_{\{u,v\} \in e^2} \left(X_{u,e}(t) - X_{v,e}(t) \right)^2$.
- For some constant C_0 , given an assignment rate ν

$$\mathbb{E}[\Delta \mathcal{L} | X(t)] \leq C_0 + \sum_{(v,e) \in U} X_{v,e}(t) |e| \left(\nu_{v,e} - \frac{1}{|e|} \sum_{u \in e} \nu_{u,e} \right) \leq -\varepsilon$$

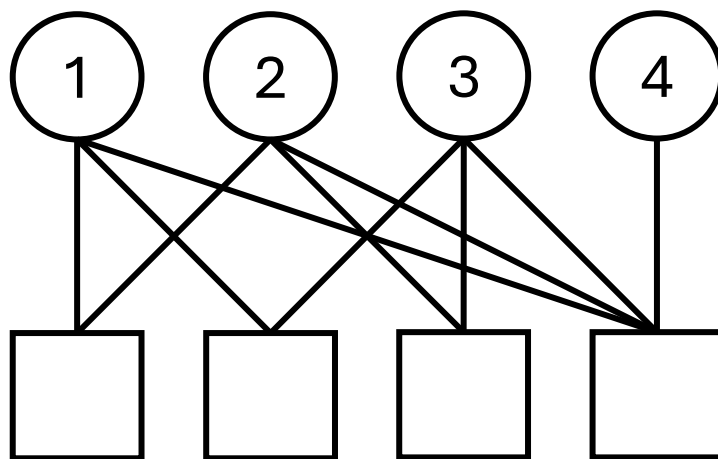
- Stability criterion gives negative drift for all $\|X(t)\|_1$ large enough.

- **Necessity:** if the criterion fails for a support U , define $Y(t) = \left(Y_{u,e}(t) \right)_{(u,e) \in U}$ by

$$Y_{v,e}(t) = \nu_{v,e}(t) - \frac{1}{|e|} \sum_{u \in e} \nu_{u,e}(t)$$

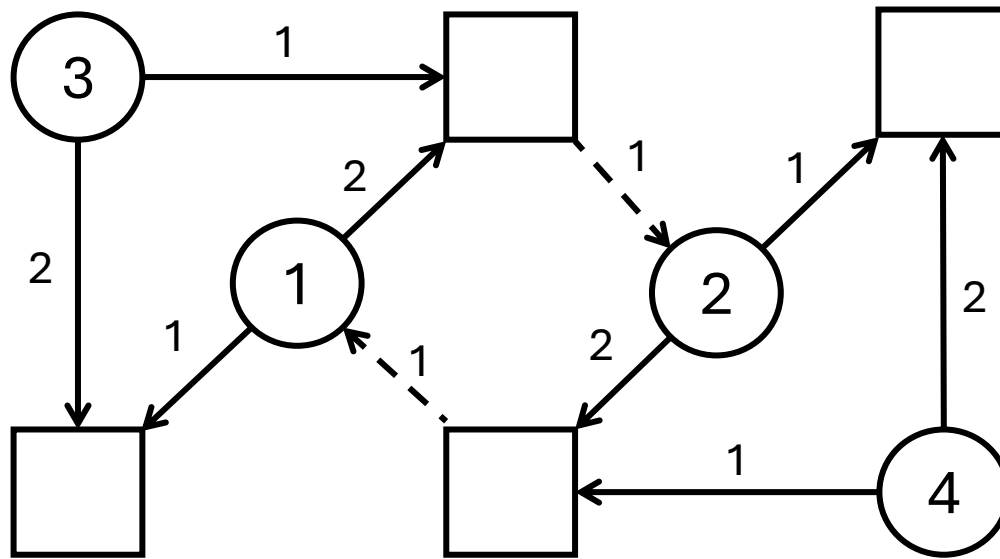
- X is a Markov chain, thus so is Y ; $X(\tau) = 0 \Rightarrow Y(\tau) = 0$.
- (G, μ) stabilisable $\Leftrightarrow \mathbb{E}[\tau] < \infty \Rightarrow Y$ positive recurrent.
- But the criterion fails, Y cannot be positive recurrent.

- Incidence matrix $A \in \{0, 1\}^{V \times E}$ of G : $A_{u,e} = 1$ iff $u \in e$.
- Stability criterion using A :
 - (Comte et al., 2021) When G is a graph, (G, μ) is stabilisable iff
 - A is surjective, and
 - There exists $\lambda \in \mathbb{R}_{>0}^E$ such that $A\lambda = \mu$.
 - (N., Bušić, 2026) The same holds when G is a hypergraph.



$$A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \det A = 2$$

- Some generalisations:
 - Non-Markovian policies;
 - Reinterpreting A as matching matrix;
 - Abandonment, service time;
 - Probabilistic matching.
- Applications:
 - Matching with multiplicities;
 - Routing, production line.



$$A = \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \end{pmatrix}$$

- Georgios Paschos

“Greedy policies are bad.”
- Georgios Paschos

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